

mcp-ratchet — Tool Accomplishment Summary & Verdict

The closing statement: what was qualified, to what level, against what requirements, what the verification concluded, the conditions on any real use, and the residual limitations carried out of the study.

Doc ID: MR-TQ-005 Rev: 2026-08-29 TQL: 5 (Criteria 3) Commit: 907c8a9 Read Order: 5 of 5

This is the tool-side equivalent of a Software Accomplishment Summary: the single document a reviewer reads to see what the qualification effort claims and on what basis. It closes the five-document study begun in MR-TQ-001. It introduces no new analysis — it states the result and the conditions attached to it.

§1 Identification

ITEM	VALUE
Tool	mcp-ratchet — runtime tool-surface drift-detection function only (MR-TQ-001 §3, §6). The prompt-injection check is explicitly outside scope.
Qualified configuration item	mcp-ratchet at git commit 907c8a9 , with FINGERPRINT_SCHEMA_VERSION = 1 .
Tool Qualification Level	TQL-5, from Criteria 3 (DO-178C 12.2.2), at any airborne software level (MR-TQ-001 §5).
Operational environment	Windows 10 (10.0.19045) 64-bit · CPython 3.13.5 · mcp 2.1.1 (MR-TQ-002 §2, as run in MR-TQ-004 §2).
Requirements	TOR-1 through TOR-11 (MR-TQ-002 §3).
Qualification data	MR-TQ-001 through MR-TQ-005, and the mcp-ratchet automated test suite at commit 907c8a9 (194 cases).

§2 What the study did

DOC	WHAT IT ESTABLISHED	OUTCOME
MR-TQ-001	Operational scenario; the 12.2.1 qualification trigger; the 12.2.2 criteria.	Trigger met (could fail to detect an error). Criteria 3 → TQL-5. Injection check bounded out.
MR-TQ-002	Eleven falsifiable Tool Operational Requirements; the operational environment; an honest gap list against the source.	TOR-9 not met (crash on a bad baseline); TOR-8 and TOR-11 partial; TOR-1–7 and TOR-10 met.
MR-TQ-003	DO-330 process areas mapped to artifacts; a committed plan to close the gaps; configuration management; the independence limitation.	Plan accepted for this study. Fix and marker work scheduled before verification.
MR-TQ-004	Requirements-based verification, one or more cases per TOR, executed on the qualified environment.	All eleven TORs PASS at commit 907c8a9 . Three commits closed the gaps. 194 tests, 0 failures.
MR-TQ-005	This summary and the verdict.	—

§3 Verification outcome

Every Tool Operational Requirement has at least one passing requirements-based verification case at commit 907c8a9 (MR-TQ-004 §3). The three requirements the source did not meet when MR-TQ-002 was written are closed by identified commits, each verified after the change:

WAS	REQUIREMENT	CLOSED BY	NOW
not met	TOR-9 — unreadable / malformed / unsupported baseline handled explicitly, no crash	78d5971	met
partial	TOR-8 — a missing baseline never reads as "no drift"; machine-readable marker	a054e4a	met
partial	TOR-11 — injection check carries a visible "not a qualified result" marker	a054e4a	met
met	TOR-1–7, TOR-10 — surface capture, canonical match, add/remove/modify classification, no cosmetic suppression, determinism, transparency	re-verified; TOR-4/5 gap cases added in 907c8a9	met

No open defect stands against any TOR.

§4 Verdict

VERDICT

On the evidence in this study, mcp-ratchet's deterministic tool-surface drift-detection function, at commit 907c8a9 , meets Tool Operational Requirements TOR-1 through TOR-11 in the operational environment of MR-TQ-002 §2, and is a defensible candidate for qualification at **TQL-5 under Criteria 3** — for the bounded use described in MR-TQ-001 §3 and §5: an *additional* automated configuration-integrity control on a separately-credited AI assistant, with human review of every drift alert and independent re-review of the assistant's interface retained on the program's normal configuration-management cadence.

The verdict holds only while all of the following hold:

CONDITION	IF IT DOES NOT HOLD
Criteria 3 usage — mcp-ratchet is added on top of existing CM, not used to eliminate a manual interface re-review.	Using it to replace a manual review is Criteria 2 — TQL-4 at Levels A and B. Not worked in this study; needs its own plan and sign-off (MR-TQ-001 §5).
The operational environment matches MR-TQ-002 §2 exactly (OS, Python, SDK, schema version, commit).	Any row change invalidates the result until re-verified (MR-TQ-004 §6).
The baseline file is held under the program's configuration management with a recorded SHA-256; a re-baseline is a separately authorised, logged act.	mcp-ratchet trusts the baseline file it is given — an unmanaged baseline is an unqualified input (MR-TQ-002 §2, MR-TQ-003 §6).
The prompt-injection check is treated as carrying no compliance credit.	Any credit taken for it is outside this qualification (TOR-11).

THE ONE CONDITION THIS STUDY CANNOT CLOSE ITSELF

The verification in MR-TQ-004 was performed by the tool's author. DO-330 expects the tool's qualification verification to be independent of its development (MR-TQ-003 §7). Before this material is used on a real program, a reviewer independent of mcp-ratchet's development must repeat or confirm the MR-TQ-004 verification against commit 907c8a9 . Until that is done, the verdict above is a candidate, not a completed qualification.

§5 Residual limitations carried out of the study

- The ceiling on what this buys (MR-TQ-001 §7).** A qualified "no drift" result means the assistant's advertised tool interface has not changed since it was qualified. It says nothing about whether the assistant's judgments are correct, it does not catch a change in model behaviour that leaves the interface untouched, and it is not a reason to skip independent human review of the assistant's output.
- Surface coverage (MR-TQ-002 §5).** The fingerprint captures the client-visible fields of the MCP Tool object. It does not capture server resources, prompts, server-level instructions, or the content of tool-call results — any of which can also steer a model. Extending the fingerprint to those is real, scoped future work; it is not done here and not claimed.
- SDK-version sensitivity.** Qualified against mcp 2.1.1. A future SDK that adds a client-visible field to the Tool model requires a TOR-1 update and re-verification. The 2.0.0 → 2.1.1 move was checked and left the field set unchanged (MR-TQ-004 §6).

§6 Configuration management record

ITEM	IDENTIFICATION	STATUS / ACTION
Qualified tool	mcp-ratchet @ 907c8a9 , FINGERPRINT_SCHEMA_VERSION = 1	action tag this commit in the repository as the qualified commit. Not yet done — the repository is currently local-only; the tag is applied when a hosting/CM decision is made.
Baseline file	baselines/<slug>.json for the credited assistant, plus a recorded SHA-256	action created and hashed at the point a real program adopts this control. Not an artifact of this study.
Qualification data	MR-TQ-001 – MR-TQ-005; the test suite at 907c8a9	Revisions are additive; a superseded revision is retained. The Read Order tags are updated across the set if a document is inserted.

§7 Closing statement

This study does not show that AI tools can be safely qualified for DO-178C credit. It shows that one specific, deterministic guardrail around such a tool — a fingerprint-and-compare check on the tool's advertised interface — can itself be qualified at the lowest tool qualification level, cheaply, against falsifiable requirements, with an honest account of the three source defects that had to be fixed to get there. The guardrail's value is bounded: it holds the edges of an existing qualification stable; it does not establish one. The single item left open is independent verification of the results in MR-TQ-004.